SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Programme: B.Tech/ MBA Tech (Computer) Year: II Semester: III

Academic Year: 2019-20

Subject: Object Oriented Programming

Date: 16 November 2019

Marks: 100
Time: 2.00 pm - 5.00 pm
Durations: 3 Hrs
No. of Pages: 2

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Final Examination (2019-20)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) In all ___5_ questions to be attempted.
4) All questions carry equal marks.
5) Answer to each new question to be started on a fresh page.
6) Figures in brackets on the right hand side indicate full marks.
7) Assume suitable data if necessary.

Q1. a. Can you achieve multiple inheritance using classes in Java? If not, how can it be achieved, explain with the help of suitable example code? [6]

b. State the Model View Controller in Java with help of a suitable diagram. [6]

c. State the use of final keyword in Java. [4]

d. What are the key differences between Abstraction and Encapsulation. [4]

Q2. a. Explain with suitable example how can you perform Abstraction by parameterization and specification. What are the advantages of each? [10]

b. i. A user-defined package 'DemoPackage' contains a class named 'Amount' which accepts a user's name and balance amount and also displays a person's details. Create another class 'Pack_use' that makes use of the above package and accepts three entries by the user.
ii. Does importing a package imports the sub packages as well? E.g. Does importing com.bob.* also import com.bob.code.*? Give appropriate reasons for your answer. [10]

Q3. a. Differentiate between Method Overloading and Method Overriding. Also give a suitable example code. [10]

b. Create a 'student' base class that accepts a student's name and roll no.; create another class 'student_A' that inherits 'student' class and accepts a student's marks in two subjects; create another class 'student_result' that calculates the total marks of the student. Display the student's details and the total marks as the output. [10]

Q4. a. Show with the help of a program how iterators can be used in Collections framework. What are the important methods declared by the iterator interface? [10]

b. What are the advantages of using Generics in Java. Also specify through a suitable program how does it provide type-safety? [10]

Q5. a. What are bounded type parameters? State their use with the help of a suitable example. [10]

b. State what are the differences between Unit testing and Integration testing. [10]

Q6. a. State what are the significant differences between an Abstract class and an Interface. [10]

b. What are containers in Java? State the use of JFrame and JApplet with the help of suitable program. [10]

Q7. a. State the use of ArrayList class with the help of a suitable program. [6]

b. Explain the what are Bounded Wildcards? Explain any of its types with the help of a suitable example. [6]

c. Write a short note on the key features of SCALA. [4]

d. Write a short note on Software Development Life Cycle. [4]
